Application of Randomized Complete Block Design in Agriculture: Case Study of CIMMYT El Batan Trial

Marcell Prasetyo
BINUS University
Jakarta, Indonesia

Ricardo Xavier Rinzi
BINUS University
Jakarta, Indonesia

Anthony
BINUS University
Jakarta, Indonesia

Aaron Jeremy Alexander
BINUS University
Jakarta, Indonesia

Satya Darma Padmakumara
BINUS University
Jakarta, Indonesia

I Made Rayxelvio Erawan Rustiadi
BINUS University
Jakarta, Indonesia

***Abstract* —** *This study evaluates the yield performance of various wheat genotypes using a Randomized Complete Block Design (RCBD). By grouping genotypes into blocks, the research aims to separate genetic potential from environmental factors. Analysis of variance confirms that the wheat variety significantly influences yield, allowing for the identification of superior genotypes. While the data largely follows standard statistical requirements, diagnostic checks reveal the presence of spatial autocorrelation, where neighboring plots influence each other's performance. Post-hoc analysis successfully categorizes the genotypes into performance groups, identifying the most productive varieties for breeding. The findings suggest that while RCBD is effective for variety selection, future agricultural experiments should consider advanced spatial modeling to better account for environmental noise in field data.*

Keywords: *Randomized Complete Block Design, wheat genotype performance, statistical analysis*

I. INTRODUCTION

Experimental design is a way to observe data systematically. It aims to find out the reasons why these results can occur. This is conducted by intentionally changing specific factors which would establish cause and effect relationships that move beyond simple correlations (Montgomery, 2017). This approach dates back to Ronald A. Fisher, who noticed that the bigger picture is often overlooked using “one factor at a time” methods. As a result, he introduced three core principles of experimental design which are randomization, replication and blocking for local control. These principles ensure that the leading cause of the results isn’t lost due to the noise of the real world (Fisher, 1935; Box, Hunter, & Hunter, 2005).

Ideally, every experimental unit should be identical. However, this is not the case in the real world. Variables that are not the main target to the experiment may affect the response. These variables are called nuisance variables and may skew results of the experiment. Therefore, a design called Randomized Complete Block Design or RCBD is applied. Two of the three main principles are used here: randomization and blocking. Randomization reduces or guards against the effect of any lurking nuisance. Meanwhile, blocking is done to systematically eliminate the effects of nuisance variables during statistical analysis (Montgomery, 2017). The combination of these two creates a strong experimental design that is able to counteract this problem.

The use of randomized complete block design is widely apparent in the field of agricultural research. Differences in the soil, moisture or even sunlight across the experimental field may affect the final result if not handled properly. In order to counteract this problem, researchers apply RCBD design to divide the experimental field into different blocks. The aim of this blocking is to ensure that every treatment experiences the same environmental changes (Gomez & Gomez, 1984). Hence, RCBD provides reliable and robust data that agricultural researchers could rely on to manage resources effectively (Box, Hunter, & Hunter, 2005). The aim of this study is to showcase the application of RCBD using a case study on the CIMMYT El Batan Trial.

II. THEORY

A. *Randomized Complete Block Design Model*

Randomized Complete Block Design (RCBD) is a common statistical design used for various agricultural experiments. The experimental variables are grouped into different blocks in order to control the nuisance variables. The fundamental idea is that the variability within a block is smaller than variability between blocks. Therefore, allowing for a more precise estimate of treatment effects.

In this study, the column position acts as the blocking factor to account for spatial heterogeneity such as soil gradients. The statistical linear model for an RCBD with a treatments and b blocks is expressed as:

$$y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}$$

Where:

- y_{ij} is the observation for the i-th treatment in the j-th block.
- μ is the overall mean.

- τ_i is the effect of the i-th treatment, for i = 1, 2, ..., a.
- β_j is the effect of the j-th block, for j = 1, 2, ..., b.
- ε_{ij} is the random error term, assumed to be independent and normally distributed N(0, σ^2)

The model assumes that the treatment and block effects are additive and that there is no interaction between treatments and blocks.

B. *Analysis of Variance (ANOVA)*

Analysis of Variance (ANOVA) is the primary statistical method used to test the hypothesis that the treatment means are equal. In an RCBD context, ANOVA partitions the Total Sum of Squares (SST) into three distinct components: the Sum of Squares of Treatments (SSTreatments), the Sum of Squares of Blocks (SSBlocks), and the Sum of Squares of Error (SSE).

The fundamental identity of the ANOVA for RCBD is:

$$SS_T = SS_{Treatments} + SS_{Blocks} + SS_E$$

Mathematically, these are calculated as:

$$SS_T = \sum_{i=1}^{a}\sum_{j=1}^{b} y_{ij}^2 - \frac{y_{..}^2}{N}$$

$$SS_{Treatments} = \frac{1}{b}\sum_{i=1}^{a} y_{i.}^2 - \frac{y_{..}^2}{N}$$

$$SS_{Blocks} = \frac{1}{a}\sum_{j=1}^{b} y_{.j}^2 - \frac{y_{..}^2}{N}$$

Where $y_{i.}$ is the total of the i-th treatment, $y_{.j}$ is the total of the j-th block, $y_{..}$ is the grand total, and N is the total number of observations (N = a X b).To test the null hypothesis (H_0: $\tau_i = 0$ for all i), we utilize the F-statistic, which compares the variance explained by the treatments to the variance attributed to random error (Mean Square Error).

$$F_0 = \frac{MS_{Treatments}}{MS_E}$$

If the calculated F_0 is greater than the critical value $F_{\alpha, a-1, (a-1)(b-1)}$, or if the p-value is less than the significance level α = 0.05, H_0 is rejected and it can be concluded that there is a significant difference between the treatments.

C. *Model Assumptions*

The validity of the statistical inferences derived from the Randomized Complete Block Design (RCBD) relies fundamentally on the properties of the experimental error terms. The linear model described in the previous section assumes that the residuals, denoted as ε_{ij} , satisfy the IIDN conditions: they must be Independent, Identically Distributed, and Normally distributed.

For All statistical tests in this study, including the main ANOVA F-test and the assumption checks, we employ a significance level of α = 0.05. The decision to reject or fail to reject the null hypothesis H_0 is determined by the p-value:

- If p-value < 0.05 : H0 is rejected (The result is statistically significant).
- If p-value > 0.05 :H0 is failed to be rejected (The result is not statistically significant).

1) Assumptions of Normality

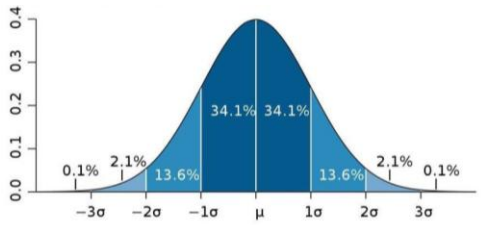


Fig.1. The theoretical normal distribution of residuals (Bell Curve)

The F-test in ANOVA is derived based on the assumption that the residuals follow a normal (Gaussian) distribution. Mathematically, this assumption is expressed as:

$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

where εij represents the random error of the i-th treatment in the j-th block. This implies that the errors have a mean of zero and a constant σ^2 variance across all treatment groups. To ensure the robustness of the Analysis of Variance (ANOVA) results, these assumptions must be rigorously verified using specific statistical tests. Durbin Deviations from normality, particularly in smaller sample sizes, can inflate the Type I error rate.

To verify this, we employ the Shapiro-Wilk test, which is preferred over the Kolmogorov-Smirnov test for its higher statistical power in detecting non-normality in ANOVA residuals.

- Hypothesis: The null hypothesis (H_0) posits that the residuals are drawn from a normally distributed population.
- Decision: A p-value greater than $\alpha = 0.05$ indicates that the null hypothesis is rejected. This confirms that the normality assumption holds. Deviations from normality can inflate the Type I error rate, making this check critical for small sample sizes.

2) Assumption of Homogeneity of Variance

Also known as homoscedasticity, this assumption requires that the variance of the response variable (Yield) remains consistent across all treatment groups (Wheat Varieties). If the variances differ significantly (heteroscedasticity), the standard error of the difference between means will be biased, rendering the F-test unreliable. To assess this, we utilize Bartlett's Test.

- Hypothesis:The null hypothesis states that the variances of the k treatment groups are equal $(H_0 = \sigma_1^2 = \sigma_2^2 = \ldots = \sigma_k^2)$.
- Decision: If the p-value obtained is greater than 0.05, H_0 is rejected and it can be concluded that the variances are homogenous which satisfies the assumption.

Bartlett's test is selected for this study due to its sensitivity and suitability for data that has already satisfied the normality assumption.

3) Assumption of Independence

The assumption of independence is arguably the most critical condition for the validity of the ANOVA model. It requires that the error term for any single observation be statistically independent of the error terms for all other observations. In the context of agricultural field experiments, this assumption is frequently challenged by spatial autocorrelation, where the yield of a specific plot is positively correlated with the yields of adjacent plots due to underlying environmental gradients, such as soil fertility clusters, moisture levels, or pest infestations.

If this assumption is violated (i.e., errors are correlated), the standard error estimates for treatment differences will be biased, leading to inflated t-statistics and a higher probability of committing a Type I error (declaring a significant difference when none exists). To verify independence, we utilize the Durbin-Watson test, which specifically detects first-order autocorrelation in the residuals.

- Hypothesis: The null hypothesis assumes no autocorrelation exists ($H0 : \rho = 0$)
- Decision: While the test statistic d (ranging from 0 to 4) provides a heuristic, modern software generates a specific p-value. A p-value less than 0.05 leads to the rejection of H0, indicating the presence of significant autocorrelation which may require spatial modeling to correct.

D. Post-Hoc Analysis

When the null hypothesis of the ANOVA F-test is rejected, post-hoc analysis is required to identify which specific treatment means differ. Since the ANOVA only indicates the presence of at least one significant difference among treatments, additional pairwise comparisons are necessary to determine the exact sources of variation among the wheat genotypes.

Fisher's Least Significant Difference (LSD) test is used for comparisons of the treatments. The LSD test compares the absolute mean difference between two treatment means against a critical value. This value is derived from the pooled error variance of the ANOVA model. A pair of treatment means is considered significantly different if their absolute difference exceeds the LSD value at a significance level of α=0.05. The formula for the LSD statistic is as follows:

$$LSD = t_{\alpha/2,df} \sqrt{MSE \cdot \left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

Where:

- $t_{\alpha/2,df}$ is the t-value using α and the degree of freedom of error
- MSE is the mean squared error
- n_i is the number of observations in i-th level
- n_j is the number of observations in j-th level

Although Fisher's LSD provides high sensitivity and greater statistical power, it does not strictly control the family-wise error rate when many comparisons are performed. Therefore, the interpretation of results must be made cautiously, especially given the large number of genotypes analyzed. The detailed results of the LSD comparisons and the resulting treatment groupings are presented and discussed in Section IV.

III. CASE STUDY

To show the applications of RBCD in a practical case study, the following dataset based on wheat varieties will be used and analyzed. It will discuss the dataset used and its justification of using RCBD.

A. *Dataset Explanation*

The dataset is based on an experiment on wheat at El Batan, Mexico by the International Center for Improvement of Maize and Wheat or CIMMYT (Besag and Higdon, 1999). It contains 4 variables which are yield, gen, col, and row. The yield variable simply denotes their ability to produce seeds. The gen variable denotes the genotype of wheat. There are 50 different varieties or genotypes of wheat which are the treatment for this case. The col and row variable denote the column and row of the observation. Each observation is assigned to a column and row number. The column number denotes in which block the observation is in, while the row number simply denotes which row the observation is in. There are 3 blocks in the dataset and 50 rows. From the 50 varieties, each of them is replicated three times and is separated into three blocks. There is no duplicate genotype in each block. Each genotype is then randomly placed in their blocks. Therefore, the total number of observations is 150. Below are the first 6 rows of the dataset.

I. WHEAT VARIETIES DATASET

Yield	Gen	Col	Row
3.37917	G01	1	1
3.58748	G02	1	2
4.32812	G03	1	3
4.53642	G04	1	4
2.59224	G05	1	5

Fig. 2. Variables in the dataset

B. *Justification of RCBD*

The goal for the case study will be to analyze whether different genotypes produce different yields. The blocking is done by the col variable. Each observation is set in its own field, which is denoted by its column. The column acts as its own homogenous block for the treatments. The treatments are randomly assigned within each block. In addition, the frequencies of each treatment in each block are only one. This ensures that genetic influence is isolated from environmental variables. Therefore, a randomized complete block design is suitable to be used for this dataset.

IV. RESULTS & DISCUSSION

A. *Analysis of Variance*

An ANOVA test is done to see whether there are any differences between the genotypes. These are the hypotheses for the test.

H0: $\tau_1 = \tau_2 = \ldots = \tau_{50}$

H1: At least one pair of τ_i ***is different***

II. ANOVA TABLE

Source	SS	df	MS	F	P-value
Gen	45.667711	49	0.9319941	1.808639	0.006611
Col	1.169396	2	0.5846979	1.134672	0.325715
Error	50.499531	98	0.5153013	-	-
Total	97.336638	149	-	-	-

Fig. 3. Analysis of variance table for the dataset

From the ANOVA table, the p-value for treatment is **0.00661**. Using an α value of 0.05, the p-value is less than α. Therefore, H0 is rejected and it can be concluded that there is at least one pair of treatments that are significantly different. Fisher's LSD test will be conducted later to further determine the different groups.

B. *Assumption Testing*

The ANOVA assumptions about the residuals will be tested next.

1. *Normality*

The first assumption to be tested is the normality of errors. A Shapiro-Wilk test will be conducted for this assumption test. Below is the hypothesis.

H0: The errors are normally distributed

H1: The errors are not normally distributed

The resulting p-value for the test is **0.06999**. Using an α value of 0.05, the p-value is more than the α value. Therefore, H0 is not rejected, and it can be concluded that the errors are normally distributed. Thus, the normality assumption is fulfilled.

2. *Homogenity*

The next assumption to be tested is the homogeneity of variances. A Bartlett test will be conducted for this assumption test. Below is the hypothesis.

H0: The variance of errors is equal

H1: The variance of errors is not equal

The resulting p-value for the test is **0.28203**. Using an α value of 0.05, the p-value is more than the α value. Therefore, H0 is not rejected, and it can be concluded that the variances of the errors are homogenous. Thus, the homogeneity assumption is fulfilled.

3. *Independence*

The last assumption to be tested is independence from the errors. A Durbin-Watson test will be conducted for this assumption test. Below is the hypothesis.

H0: The errors are not autocorrelated

H1: The errors are autocorrelated

The resulting p-value for the test is **3.55e-12**. Using an α value of 0.05, the p-value is less than α. Therefore, H0 is rejected and it can be concluded that the errors are autocorrelated. Thus, the independence assumption is violated.

The existence of this violation suggests the presence of spatial autocorrelation where the yield is influenced by neighboring soil. This phenomenon is well-documented in various agricultural literature, most notably by Besag and Higdon (1999) who use this dataset to propose a more spatial model over traditional RCBD analysis.

C. *Fisher's LSD*

This test is to determine which pairs of genotypes are significantly different from each other. There are 50 genotypes and the mean of each will be calculated and compared to each other. For simplicity, the genotype and their means are ordered from greatest to least. Then, the absolute mean difference between two genotypes will be compared to the LSD test value. The value for this LSD test is **1.163**. If the absolute difference between the two genotypes is more than the LSD value, then the two genotypes are significantly different. Below is a condensed table showing the top 3 largest means and top 3 smallest means.

III. GENOTYPE GROUPS

Gen	Mean Yield	Groups
G17	3.610620	a
G04	3.564330	ab
G01	3.402317	abc
...	...	...
G33	1.712730	ij
G30	1.604720	ij
G36	1.465850	j

Fig.4. Comparisons of groups based on LSD test

From the table, genotypes are divided into groups. Genotypes within the same group have no significant difference between their means. Meanwhile, genotypes not in the same group have a significant difference between their means. For example, G17 is in group A, therefore every genotype in group A has no significant difference between their means with G17. On the other hand, G33, G30, and G36 are not in group A. Therefore, their means are significantly different from G17. The full table can be seen in the APPENDIX I.

V. CONCLUSION

The CIMMYT wheat dataset study shows the advantages and challenges of using a Randomized Complete Block Design (RCBD) in agricultural research. The experiment separated genetic performance from wide environmental variables by grouping the 50 wheat genotypes into blocks based on column positions. We can reject the null hypothesis and continue to determine the most productive varieties since the results of the ANOVA test proved ($p = 0.0066$) that wheat yield is highly influenced by genotype.

Fisher's Least Significant Difference (LSD) was used to statistically group the genotypes. Based on the post-hoc test results, types like G17 and G04 represent the top class of seed output, while others like G36 fell well short. These groups offer farmers and breeders an accessible, data-driven method for selecting which genotypes to prioritize for future harvests.

However, an important understanding of the concept of field experiments was shown by the statistical "review" of the model. The Durbin-Watson test showed an important failure of independence even though the data had a normal distribution and constant variance. This suggests that "spatial autocorrelation" where the health of one sample probably influenced its neighbor through shared soil areas or moisture levels was caused by the close proximity of plots in the field even with blocking.

In summary, autocorrelation indicates that agricultural data is rarely fully accurate even if the RCBD proved to be a reliable method to find noticeable differences across wheat types. The highest-yielding genotypes are successfully found in this study, but it also shows that using advanced spatial modeling might help better distinguish the "signal" of genetic potential from the "noise" of the Mexican soil in future experiments on this specific location.

REFERENCES

1. Besag, J., & Higdon, D. (1999). Bayesian analysis of agricultural field experiments. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *61*(4), 691–746.
2. Box, G. E. P., Hunter, J. S., & Hunter, W. G. (2005). *Statistics for experimenters: Design, innovation, and discovery*(2nd ed.). Wiley-Interscience.
3. Fisher, R. A. (1935). *The design of experiments*. Oliver and Boyd.
4. Gomez, K. A., & Gomez, A. A. (1984). *Statistical Procedures for Agricultural Research* (2nd ed.). John Wiley & Sons.
5. Montgomery, D. C. (2017). *Design and analysis of experiments* (9th ed.). John Wiley & Sons.

VI. APPENDIX I: FISHER LSD TABLE

Below is the full table for each comparison of genotype. The mean yield and groups are listed as such.

IV. COMPLETE GENOTYPE GROUPS

Gen	Mean Yield	Groups
G17	3.610620	a
G04	3.564330	ab
G01	3.402317	abc
G32	3.294307	abcd
G20	3.263447	abcde
G11	3.255730	abcde
G09	3.224870	abcdef
G02	3.093717	abcdefg
G18	3.062857	abcdefg
G03	3.047427	abcdefg
G12	3.039710	abcdefg
G42	2.900840	abcdefgh
G16	2.893127	abcdefgh
G08	2.754257	abcdefghi
G14	2.731110	abcdefghi
G41	2.669390	abcdefghi
G50	2.669390	abcdefghi
G39	2.638530	abcdefghi
G10	2.592240	abcdefghij
G15	2.561380	abcdefghij
G44	2.561380	abcdefghij
G06	2.530520	abcdefghij
G40	2.530520	abcdefghij
G31	2.445657	bcdefghij
G26	2.422510	bcdefghij

Fig.5. Complete comparisons of genotypes

V. COMPLETE GENOTYPE GROUPS (CONT.)

Gen	Mean Yield	Groups
G43	2.399367	cdefghij
G22	2.345360	cdefghij
G21	2.329930	cdefghij
G05	2.299070	cdefghij
G07	2.268210	cdefghij
G13	2.252780	cdefghij
G25	2.252780	cdefghij
G47	2.221920	defghij
G34	2.113910	efghij
G19	2.083050	fghij
G24	2.083050	fghij
G29	2.083050	fghij
G37	2.044477	ghij
G35	1.990470	ghij
G23	1.851600	hij
G27	1.805310	hij
G38	1.789880	hij
G48	1.789880	hij
G49	1.789880	hij
G28	1.774450	hij
G46	1.759020	hij
G45	1.743590	hij
G33	1.712730	ij
G30	1.604720	ij
G36	1.465850	j

Fig.5. Complete comparisons of genotypes (cont.)